Pooled powers and separated radicals for Diophantine consistency

Research prepared for Vladimir Reshetnikov

25 October 2026

Abstract

We construct explicit polynomial equations whose lack of solutions in their specified domains is provably equivalent in ZFC to syntactic consistency of ZFC. The principal quartic constructions combine pooled power certificates, exact coordinate elimination, and reversible changes of residual basis to reduce their numbers of quantified coordinates. A separate four-radical construction combines an auxiliary Pell certificate with three existing square conditions, using separated powers of three to prevent cancellation between equal irrational square classes. This provides a twelve-integer alternative with a particularly direct reduction proof; it is not a global variable-degree-operation record. An inverse Lucas correction is also composed with an unsquared Pell trace to shorten a ten-positive-variable evaluator. Every result is supplied with literal equations, fixed coefficients, ordinary proofs, constructive maps, and reproducible finite checks. Inherited logical calibration remains explicit. No consistency decision, global optimum, or new end-to-end kernel certificate is claimed.

Contents

1 The objective and the comparison snapshot **2**

2 Results and tradeoffs **2**

3 A four-root elimination in twelve unrestricted integers **3**
3.1 Complete polynomial and tuple . . . 4
3.2 The four-root identity and denominator clearing . . . 4
3.3 Separation and rational Pell integrality . . . 5
3.4 Total semantic maps . . . 5
3.5 Exact degree and reproducibility . . . 6

4 Natural quartics: shared syntax wires and pooled powers **6**
4.1 Three exact syntax-wire savings . . . 7
4.2 A doubled lower Pell frame at no coordinate cost . . . 8
4.3 A three-coordinate natural pool . . . 8
4.4 The orientation issue and its exact repair . . . 9
4.5 Completeness, logical inheritance, and finite evidence . . . 10

5 Strict-frame pooling: a direct quartic in 217 integers **10**
5.1 The unchanged frame supports both power pools . . . 11
5.2 The odd-linear beta bootstrap in the correct order . . . 12
5.3 Completeness and unchanged arithmetic budgets . . . 13
5.4 Exact maps, complete coefficients, and evidence . . . 13

6 An inverse correction composed with the explicit Pell trace **13**
6.1 Complete equation . . . 13
6.2 Signed soundness and the corrected index . . . 14
6.3 Project before recovering the full denominator . . . 15
6.4 Strictly positive completeness . . . 15
6.5 Exact cost, coefficients and reproducibility . . . 16
6.6 A small admissible-denominator interpolation . . . 16

7 Verification and inherited calibration **16**

A The full original radix67 coefficient tuple **17**

B The full one anchor7309 coefficient tuple **19**

1 The objective and the comparison snapshot

For each specified domain D and fixed polynomial P the target is

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

Our proofs establish solvability equivalent to existence of a finite syntactic refutation. Neither consistency nor unsolvability is separately asserted. The domains $\mathbb{Z}$, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0}$ are part of the theorems and may not be silently interchanged.

The comparison snapshot is `892d73e1ab4bc5fd34ecca1967b06dc47ca95494` of ProveIt. It includes the latest pooled-power, quadratic-projection, strict-horizon, trace-radical and linear-CRT constructions, as well as our preceding report. We compare with actual supplied equations rather than only earlier headline records. The accompanying baseline audit records sources and exact metrics.

A bounded late comparison at `fb1bed5eb7196cf35abfaa746c18c6ee7eae6cfb` found no newly supplied endpoint dominating the new quartic dimensions or ten-positive circuit tradeoffs. Its display had moved to the credited zero-weight6007 tuple: 15 positive variables, degree 157, 66 integer-leaf operations and 141,226 raw coefficient bytes. Our checked one-anchor7309 sidecar remains a deliberately frozen earlier specialization, not a claim to the latest storage record; no new tuple's calibration is silently substituted.

Operation counts use live binary addition, subtraction and multiplication over integer leaves. Squares and fixed scalar multiplications each cost one; register reuse is free. Fixed coefficient ports are not variables, but their entire coherent integer tuples are part of every equation. Rational-leaf counts use a different convention and are not silently compared. Exact total degree, collected support, coefficient height and operation count measure different costs; a smaller-variable equation need not improve every measure.

2 Results and tradeoffs

The new integer equations compose credited incoming constructions rather than attributing their component lemmas anew. The strongest incoming pooled-power dimension endpoint has 225 variables, 162 rows, 7,867 terms and height 128,446. The new 217-variable equations have more terms but fewer variables and rows and much smaller height. The immediate strict- horizon parent has 235 variables; the matched reduction saves eighteen.

Direct quartic over	Variables	Rows	Terms	Height
$\mathbb{Z}$, low height	217	154	25,949	4,576
$\mathbb{Z}$, fewer terms	217	154	14,615	16,866
$\mathbb{Z}$, low height	218	155	15,631	4,576
$\mathbb{Z}$, fewer terms	218	155	8,817	16,866
$\mathbb{N}$, fewer terms	202	154	9,818	239,204,765
$\mathbb{N}$, lower height	202	154	12,706	31,145,294
$\mathbb{N}$	203	155	6,834	228,943,870
$\mathbb{N}$	204	156	6,756	225,656,347

Table 1: Literal degree-four equations in this release. Each has all stated coordinates live. Height is the largest absolute collected coefficient.

The natural parent has 206 variables. Its canonical-sign construction is retained, while three syntax coordinates are eliminated and a shared power block saves one further coordinate. The displayed support and height tradeoffs are obtained by explicit reversible residual shears; no search optimality is asserted. The lower-dimensional equations do not uniformly improve every older support or height endpoint.

Domain	Variables	Degree	Operations	Matched comparison
$\mathbb{Z}_{>0}$	10	232	85	Incoming 232/87
$\mathbb{Z}_{>0}$	10	348	77	Between 352/76 and 290/79
$\mathbb{Z}$	12	512	132	Tradeoff with 500/139

Table 2: Small-variable polynomial circuits with integer leaves. In the last column pairs are degree/operations, with variable count fixed.

The twelve-integer equation is included for its direct four-root reduction and its within-twelve-variable tradeoff, not as a global frontier record. It saves seven operations against the existing degree-500, 139-operation form but raises the degree by twelve. Existing eleven-integer equations, including degree 398 at 121 operations, dominate its variable-degree-cost triple. The preceding thirteen-integer degree-152, 119-operation form remains a different low-degree tradeoff.

The complete one-anchor7309 coefficient specialization is credited to its incoming source. Its four ports use 172,011 raw bytes. Applying a proved uniform front to this coherent tuple is separately checked and does not identify its specialized polynomial with that of the older radix67 tuple.

3 A four-root elimination in twelve unrestricted integers

Result and comparison. The literal equation `integer12_degree512.json` has twelve independent integer unknowns, exact total degree 512, and 132 live binary operations (82 multiplications, 25 additions, 25 subtractions). Squares and scalar multiplications are charged. This is a new witness-elimination proof and a fixed-twelve-variable tradeoff, not a variable-count or degree record. Against the catalogued 12/500/139 point it saves seven operations at twelve more degrees; against 12/1084/123 it saves 572 degrees at nine more operations. The catalogue already contains the eleven-integer point 11/398/121, which dominates it across all three metrics. The retained 13/152/119 remains a separate lower-degree tradeoff. All triples here are variables/degree/operations.

The ordinary signed kernel and its native/compiler/syntactic-refutation calibration are inherited explicitly, not reproved by a finite checker. Composing that calibration with the total zero-set maps below proves in ordinary ZFC that this polynomial's universal integer nonvanishing is equivalent to syntactic Con(ZFC). No consistency assumption, consistency proof, new end-to-end proof-assistant certificate or global minimum is claimed.

3.1 Complete polynomial and tuple

The unrestricted variables, in literal order, are

$$j, C, z, h, r, v, w, T, c, n, p_1, p_2 \in \mathbb{Z}.$$

The six complete fixed integers are $\beta, K_0, M_C, M_F, A_{\text{cone}}, B_{\text{cone}}$, where M_F is already shifted, $\beta = 2^{1064921} - 1$, $A_{\text{cone}} = 2^{1113}$ and $B_{\text{cone}} = 2^{666104}$. All six byte payloads and their finite construction are supplied under `inherited_round20/inherited/coefficients`; no coefficient port is left unspecified. Define polynomial abbreviations

$$\begin{gathered}
J = 6j + 1, \quad q_- = \beta J, \quad q = q_- + 1, \\
W = 12w + 4, \quad X = qW, \quad Y = v, \\
F = (K_0 + W)C - zq_-, \quad R = [q(q - F) - C](q^2 - 1) + (M_C + qM_F)J, \\
E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \\
\ell = EY, \quad k = R + 1 + h\ell, \quad \eta = c - Yk, \quad L = \ell k, \\
I = (2r + 1)c^2, \quad D_s = \Delta I^2, \quad F_s = 1 + D_s, \quad Q = \Delta D_s, \\
V = F_s(cT - R) - c, \quad q_3 = q^3, \quad M = 2q_3, \quad U = MH, \\
K_{\text{lo}} = X(2X - 3c) - 2, \quad D_{\text{err}} = (A_{\text{cone}}F - q)^2 + (B_{\text{cone}}C - q)^2, \\
B_0 = (Yk)^2(q^2 - D_{\text{err}}) - q^2[4(Y\eta)^2 + (kq^2)^2], \quad G = 4(B_0 - p_1^2 - p_2^2) - 3, \\
D_t = 1 + L(L + k), \quad U_2 = U^2, \quad Z_c = U(n + 2L) + MK_{\text{lo}} + H(Y - q_3), \\
\delta = Q - 1, \quad C_y = QV^2 - 1, \quad D_U = \delta U_2.
\end{gathered}$$

The five inputs to a homogeneous norm are

$$S = \delta Z_c^2, \qquad A = D_U F_s, \qquad B = 9D_U D_t, \qquad C_* = 729 D_U G, \qquad D = 81 U_2 C_y.$$

The following schedule defines the entire output:

$$\begin{gathered}
s_+ = S + D, \quad a_+ = A + C_*, \quad a_\times = AC_*, \\
e_2 = a_\times + Ba_+, \quad m = s_+ - a_+ - B, \quad m_2 = m^2, \\
d_2 = SD, \quad h_2 = m_2 + 4(d_2 - e_2), \quad e_3 = a_\times B, \\
n_0 = h_2^2 + 16(d_2 m_2 - 4e_3 s_+), \quad \boxed{P = n_0^2 - 64 d_2 (m h_2 - 16 e_3)^2.}
\end{gathered}$$

Every displayed register is an abbreviation. The literal DAG uses only integer leaves, addition, subtraction and multiplication.

3.2 The four-root identity and denominator clearing

Let $\mathcal{H}_4(S, A, B, C_*, D)$ denote the final schedule. It is homogeneous of degree eight in its five arguments and satisfies the formal identity

$$\mathcal{H}_4(z^2, A, B, C_*, D) = \prod_{\epsilon \in \{-1,1\}^4} (z - \epsilon_1\sqrt{A} - \epsilon_2\sqrt{B} - \epsilon_3\sqrt{C_*} - \epsilon_4\sqrt{D}).$$

This holds for arbitrary complex roots, including coincident square classes. The independent standard-library checker multiplies all sixteen factors exactly and compares all 495 coefficients.

Unconditionally, q is even and nonzero, $|X| \geq 8$, and $a = (X+1)Y$ is zero or has absolute value at least seven. Thus $\Delta = (a+2)^2 - 1 \geq 3$, H is odd and nonzero, and $U \neq 0$. Moreover $D_t, F_s \geq 1$ because $D_t = 1 + \ell(\ell+1)k^2$ and ℓ is even. As $Q = (\Delta I)^2$, either $Q = 0$ or $Q \geq 9$; hence δ, C_y never vanish. Also $G \equiv 1 \pmod 4$, so $G \neq 0$. With $\rho = Z_c/U$, the exact proof identity is

$$P = \delta^8 U^{16} \prod_{\epsilon \in \{-1,1\}^4} \left(\rho - \epsilon_1 \sqrt{F_s} - 3\epsilon_2 \sqrt{D_t} - 27\epsilon_3 \sqrt{G} - 9\epsilon_4 \sqrt{C_y/\delta} \right).$$

The factor δ^8 is already cleared in the literal polynomial; the quotient above is proof notation, not a division gate.

3.3 Separation and rational Pell integrality

The authentic β is $1 \pmod 3$. For every integer assignment,

$$q = X = -1, \quad W = 1, \quad a = H = \Delta = 0 \pmod 3,$$

so $F_s = D_t = 1$ and $\delta = C_y = -1 \pmod 3$. For D_t, use $\ell = -Y^2$ and $Y^4 - Y^2 = 0 \pmod 3$. In the order

$$\sqrt{F_s}, \quad 3\sqrt{D_t}, \quad 9\sqrt{C_y/\delta}, \quad 27\sqrt{G},$$

the squared 3-adic valuations are $0, 2, 4, 6 + v_3(G) \geq 6$.

Write each nonzero rational radicand as du^2 with d a signed squarefree integer and $u \in \mathbb{Q}^\times$. Distinct $\sqrt{d}$ are linearly independent over $\mathbb{Q}$, by sign-change character projections in a multiquadratic extension (including $\sqrt{-1}$). Consider only an irrational class $d \neq 1$; the rational carrier belongs to class 1. Any class meeting the first three roots has $v_3(d) = 0$, and its coefficients have a unique least valuation among $0, 1, 2, \geq 3$. Their signed sum cannot vanish. A class containing only the guard cannot vanish either. Consequently every root in a vanishing norm factor is rational. This includes negative radicands and dependent square classes; it does not assume that at most one term is imaginary.

At a zero there are signed rationals f, τ, t, g such that

$$f^2 = F_s, \quad \tau^2 = D_t, \quad t^2 = C_y/\delta, \quad g^2 = G, \qquad \rho = f + 3\tau + 9t + 27g.$$

The first, second and fourth are integers, and g is odd. For the third, $QV^2 - (Q-1)t^2 = 1$. If $Q = 0$, then $t = \pm 1$. If $Q > 1$, rational Pell integrality follows by descent. Indeed, write $t = r/m$ in lowest terms; $m^2 \mid Q - 1$. After sign reflection, $u = |V| > 1$ implies $u < t < Qu/(Q-1)$. The norm-preserving map

$$u' = (2Q-1)u - 2(Q-1)t, \qquad t' = (2Q-1)t - 2Qu$$

has $u' \in \mathbb{Z} \setminus \{0\}$, $|u'| < u$, and retains the reduced denominator m because $2Q - 1 \equiv 1 \pmod m$. Descent reaches $u = 1, t = 1$, forcing $m = 1$. Thus all four extracted witnesses are integers.

3.4 Total semantic maps

In the inherited fifteen-variable kernel put $Z_o = Un_o + MK_{\mathrm{lo}} + H(Y - q_3)$. Its residuals are

$$(Z_o + Uk)^2 + 4U(LZ_o - U) = 0, \qquad f_o^2 - \Delta I^2 - 1 = 0,$$
$$V_o^2 - (Q_o - 1)y_o(2V_o + y_o) - 1 = 0, \qquad B_0 - 1 - p_1^2 - p_2^2 - p_3(p_3+1) = 0,$$

where $Q_o = \Delta(f_o^2 - 1)$ and $V_o = f_o^2(cT - R) - c$. From every new zero restore

$$f_o = f, \quad y_o = t - V, \quad p_3 = (g-1)/2, \quad n_o = n - k - f - \tau - 9t - 27g.$$

Then $Z_o/U + 2L + k = 2\tau$, and

$$(Z + Uk)^2 + 4U(LZ - U) = (Z + U(2L + k))^2 - 4U^2 D_t$$

proves the first residual. The remaining three follow from the extracted square and Pell equations. Conversely, every old zero gives the integer

$$\tau = \tfrac{1}{2}(Z_o/U + 2L + k)$$

(the rational square is $4D_t$), together with $g = 2p_3+1$ and $t = V_o+y_o$. Set $n = n_o+k+f_o+\tau+9t+27g$ and delete f_o, y_o, p_3. Then a factor of the new norm vanishes. No canonical-witness restriction or unproved quotient integrality is introduced.

3.5 Exact degree and reproducibility

Let

$$q_0 = 6\beta j, \quad a_0 = 12q_0 wv, \quad F_0 = 12wC - q_0 z, \quad R_0 = -q_0^3 F_0, \quad U_0 = 8q_0^3 a_0.$$

Also $Q_0 = 4a_0^4 r^2 c^4$ and $V_0 = -4a_0^2 r^2 c^4 R_0$. The degrees of (S, A, B, C_*, D) are bounded by $(48, 42, 48, 46, 64)$. The last is attained, with entire leading form

$$D_0 = 81U_0^2 Q_0 V_0^2 = 2^{12}3^4 q_0^6 a_0^{10} r^6 c^{12} R_0^2.$$

Homogeneity and the coefficient $[D^8]\mathcal{H}_4 = 1$ show that $[P]_{512} = D_0^8$. Every other norm monomial has strictly smaller weighted degree. In particular

$$[j^{192} w^{80} v^{80} r^{48} c^{96} z^{16}]P = 2^{448}3^{304}\beta^{192} > 0.$$

This authentic fixed-coefficient witness has total degree 512.

The portable commands `build.py -check`, `check.py`, and `reconstruct_coefficients.py` require only the Python standard library. Normal and optimized runs authenticate all twelve live variables, all six coefficient ports, the complete 495-term norm, the restore identities, the leading form, 729 modulus-three fixtures, and rejection of all 132 live-gate corruptions. All six full numerals (615,883 raw bytes) are rebuilt from the retained finite recipe. Source pins distinguish current comparison commit `892d73e1ab4bc5fd34ecca1967b06dc47ca95494` from the immediate round20 parent and its source153 signed packet. Finite checks do not replace the unbounded separation/descent proof or inherited compiler calibration.

4 Natural quartics: shared syntax wires and pooled powers

All coordinates in this section range independently over $\mathbb{N} = \{0, 1, 2, \ldots\}$. The new principal equation is

$$P_{202}(x_0, \ldots, x_{201}) = \sum_{j=0}^{153} R_j(x)^2 = 0,$$

where every R_j is the literal quadratic polynomial in `natural202/equation.json`; all collected coefficients and all rendered residuals are supplied alongside it. The complete ordinary proof is `PROOF.md`. Relative to the explicitly retained C222/ZF/ZFC calibration, ZFC proves

$$\neg\exists x \in \mathbb{N}^{202}\ P_{202}(x) = 0 \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

This is not a consistency proof, a global optimality claim, or a newly kernel-checked end-to-end formal derivation.

Endpoint	Variables	Rows	Monomials	Coefficient height
Inherited natural endpoint	206	158	8210	130686376
New dimension/support form	202	154	9818	239204765
New low-height form	202	154	12706	31145294
New sparse companion	203	155	6834	228943870
New sparse companion	204	156	6756	225656347

The four-coordinate reduction precedes optional residual-basis changes. The latter are explicit invertible integer row shears, not coordinate transformations: they preserve the common zero set on the same natural orthant. The low-height endpoint trades additional monomials for smaller coefficients. No search or optimality claim is implicit in these choices.

4.1 Three exact syntax-wire savings

Let $b = \texttt{crt.encoded.b}$, $M = \texttt{mul.36}$ and $L = 2b + M$. Three literal parent definitions share the shifted square

$$G = (2L + 6)(2L + 7) = 4L^2 + 26L + 42 :$$

$$\begin{aligned} \texttt{mul.43} &= G + 8L + 30, \\ \texttt{mul.143} &= G + 4L + 14, \\ \texttt{mul.166} &= 4G - 4L - 12. \end{aligned}$$

Replace the three coordinates by G and its one quadratic defining row. The last restoration is natural on that row because $4G - 4L - 12 = 16L^2 + 100L + 156 \geq 0$; it is not asserted to be unconditionally nonnegative on arbitrary (G, L). In the sparse parent charts the third retained coordinate is instead

$$\texttt{add.171} = 8G + 4b + 2\texttt{mul.60} + 13,$$

with the same two-coordinate saving. Conversely an arbitrary parent zero supplies the natural G by its displayed polynomial. Exact substitution turns the three old defining rows into multiples $1, 1, 4$ (or $1, 1, 8$) of the one new row. All remaining rows are preserved by substitution.

Finally,

$$\texttt{mul.366} = F(4 + 4b + 2M), \qquad \texttt{mul.370} = F(6 + 4b + 2M)$$

allow the unconditional natural restoration $\texttt{mul.370} = \texttt{mul.366} + 2F$ and deletion of its duplicate row. These changes save three genuine coordinates and rows. They leave every syntax label, local source value, and inherited numerical majorant exactly unchanged.

4.2 A doubled lower Pell frame at no coordinate cost

The retained semantic frame W is fixed after the intended powers and representatives. Write $A = \ell + \sigma$, with $\ell, \sigma \geq 0$. Its lower graph originally used

$$P = W(z_{\text{old}} + 1), \qquad R = (W+1)P, \qquad \ell^2 + P^2 - R^2 = 1.$$

Substitute $z_{\text{old}} = 2z + 1$. Thus $P = 2W(z+1) > 0$ with the same coordinates and degree. Classification of the lower Pell norm gives

$$\ell = X_e(W+1), \qquad P = Y_e(W+1), \qquad e > 0.$$

The second-order recurrence for Y_e reduces modulo $2W$ to the recurrence for e, since $2(W+1) \equiv 2 \pmod{2W}$. Hence $Y_e(W+1) \equiv e \pmod{2W}$, so $e \geq 2W$. For $W \geq 2$,

$$A \geq \ell \geq (W+1)^{2W} > 4W^{2W}. \tag{N1}$$

Indeed, the first three binomial terms of $(1+1/W)^{2W}$ already sum to $5 - 1/W > 4$. For completeness choose $e = 2W$; the quotient $Y_{2W}(W+1)/(2W)$ is a positive integer. The subsequent common-root congruence still permits choosing A arbitrarily large above ℓ. This is a cofinal auxiliary-witness strengthening, not a bijection on all old lower Pell tuples.

4.3 A three-coordinate natural pool

The three exact requests at base $n = q$ have indices and outputs

$$(k_0, k_1, k_2) = (t, c, x), \qquad c = y + 1, \qquad (m_0, m_1, m_2) = (D_t, D_y, D_x).$$

In the emitted charts each m_i is the affine sum of its pooled remainders, its pooled slack, and 1. Their natural consecutive pairs (Y_i, Z_i) retain all three norms. The old rows used a modulus coordinate $T = 2An - n^2 - 1$ and three natural quotients h_i:

$$nY_i - Z_i = m_i + Th_i. \tag{N2}$$

Delete these four coordinates and four rows. Introduce only three natural coordinates U, M_p, K with

$$\begin{aligned} U &= \sum_i Y_i^2, \\ K &= 2\sum_i Y_i(m_i + Z_i) + (2A - n)M_p - nU, \\ \sum_i (m_i + Z_i)^2 + M_p &= nK. \end{aligned} \tag{N3}$$

All three rows are quadratic. They imply the exact polynomial identity

$$\sum_i (m_i + Z_i - nY_i)^2 = TM_p. \tag{N4}$$

If their residuals are S_0, S_1, S_2 in displayed order, the left side minus TM_p equals $S_2 + nS_1 - n^2 S_0$.

For canonical true powers define

$$U = \sum_i Y_i^2, \qquad M_p = T\sum_i h_i^2,$$

and define K by the middle row of (N3). It is an integer, and the final row yields

$$nK = \sum_i (m_i + Z_i)^2 + M_p \geq 0.$$

Thus K is natural when $n, T > 0$. No signed-coordinate relaxation is hidden in the pooling construction.

4.4 The orientation issue and its exact repair

Deleting (N2) removes the sole predecessor-orientation incidence for the $c = y + 1$ seed. It is therefore invalid simply to assume all three consecutive pairs remain predecessors. Here is the noncircular repair.

The stronger frame first authenticates the complete semantic anchor. For any consecutive pair, $X = AY - Z$ satisfies $X^2 - (A^2 - 1)Y^2 = 1$. Classification applied to $(|X|, Y)$ supplies $Y = Y_j(A)$ even when the sign of X is unknown. The retained merged error-square cap depends only on these ordinates; its original small-error argument still proves $j = k$. The anchor itself retains positive ordinary coordinates and every private witness.

The t and x pairs retain their individual positive-output rows at bases $F^2 - 1$ and B, respectively. A row $bY = Z + R$ with $R \geq 0$ and $1 \leq b < A$ excludes $Y = 0$ and implies $AY - Z > 0$. Their predecessor orientations therefore survive. The three unpooled power requests recover V, N, q by the old argument, so $n = q \geq 1$ before the pool is interpreted. The unchanged literal frame bounds $n, m_i, k_i \leq W$, giving

$$T \geq 2A - W^2 - 1 > A > 4W^{2W} > 0. \tag{N5}$$

The decisive additive, rather than merely maximum, bounds are

$$c + t \leq W - 1, \qquad c + x \leq W - 1, \qquad 2c \leq W - 1, \qquad c + 1 \leq W - 1. \tag{N6}$$

The independent verifier checks all four as exact coefficientwise nonnegative differences in the literal frame. For the sparse charts it first expands the retained `add.594` row and uses the retained cofinal/floor identity $F + Fr + Fs + r = y + 1$. No factorial or power interpretation is used in checking these inequalities.

Modulo T, the standard Pell recurrence gives $X_k + (n - A)Y_k \equiv n^k$. Its conjugate is the inverse, because $(n - A)^2 \equiv A^2 - 1 \pmod{T}$. Thus the actual pair has

$$nY - Z \equiv \begin{cases} n^k & \text{if } X = X_k, \\ -n^{-k} & \text{if } X = -X_k. \end{cases} \pmod{T}$$

Here inverses exist since $T \equiv -1 \pmod{n}$. Multiply (N4)'s errors by n^c. Their ordinary integer residues are

$$\begin{aligned} &\delta_0 = m_0 n^c - n^{c+t}, && \delta_2 = m_2 n^c - n^{c+x}, \\ &\delta_1 = m_1 n^c - n^{2c} && \text{in the predecessor case,} \\ &\delta_1 = m_1 n^c + 1 && \text{in the reversed case.} \end{aligned}$$

Consequently T divides $E = \delta_0^2 + \delta_1^2 + \delta_2^2$. By (N6), each displayed power and each $m_i n^c$ is at most W^{W-1}. Therefore

$$0 \leq E \leq 12W^{2W-2} \leq 3W^{2W} < T.$$

It follows that $E = 0$. The reversed case is impossible because $m_1 n^c + 1 > 0$. Cancellation of the positive n^c now gives all three exact powers and the missing predecessor orientation.

The old quotients are genuinely natural, not merely integral: the standard positive-power identity is

$$nY_k - Y_{k-1} - n^k = T \sum_{j=0}^{k-1} n^{k-1-j} Y_j \geq 0.$$

Thus all four deleted coordinates in (N2) can be restored, as can the old frame coordinate and the syntax wires. Every parent residual, including the local nonnegative evaluation row, is recovered.

4.5 Completeness, logical inheritance, and finite evidence

Starting from a finite refutation, the parent's constructive proof chooses the even horizon, factorial, beta representative, precision-N^2 CRT lifts, canonical local signs, semantic label anchor, self-sized radix and denominators in its unchanged order. Choose the stronger auxiliary lower Pell witness only after these semantic values, then choose A in its same root-congruence class and all canonical Pell witnesses afresh. Larger A only strengthens the merged-error completeness bound. The formulas above supply $G, U, M_p, K \in \mathbb{N}$. This proves refutation-to-zero without asserting that every arbitrary old auxiliary tuple lifts.

Soundness restores the complete parent, whose retained acyclic proof then recovers the descending factorial, carry-free binomial ranges, reciprocal CRT, unchanged local source, factorial-weight Horn flow and calibrated finite syntactic ZFC refutation. No consistency or arithmetic-soundness assumption is used. The full proof closure and notices are retained, and its ordinary theorem is distinguished from the finite Python evidence.

The standard-library commands

```
python3 -B verify.py -rebuild
python3 -B -O verify.py -rebuild
```

check the pinned inherited source, every new source row, the exact natural restoration identities, the pooled identity and additive frame bounds, all inverse row shears, every collected coefficient and parsed printed residual, and byte-identical rebuilding. Deleting the source-evaluation row is an explicit rejected negative control. These exact finite audits complement, rather than replace, the ordinary unbounded proof.

5 Strict-frame pooling: a direct quartic in 217 integers

The principal equation is the literal quartic

$$P_{217}(x_0, \ldots, x_{216}) = \sum_{j=0}^{153} R_j(\boldsymbol{x})^2 = 0, \qquad \boldsymbol{x} \in \mathbb{Z}^{217},$$

where every quadratic residual and every collected coefficient is supplied in the accompanying data. In ordinary ZFC, its integer nonsolvability is equivalent to Con(ZFC), through the retained numerical C222 and finite ZF/ZFC-refutation calibration. This is not a proof of consistency, a new proof-assistant closure, or a global minimum claim.

Profile	Variables	Rows	Terms	Height
Low height	217	154	25,949	4,576
Smaller support	217	154	14,615	16,866
Low-height companion	218	155	15,631	4,576
Sparse companion	218	155	8,817	16,866

Every degree is exactly four and every residual coefficient has absolute value at most 64. These four profiles are mutually nondominated in the four displayed costs. The matched Strict Horizons parents have 235/172 and 236/173 variables/rows: each branch saves 18 of each. The distinct incoming Pooled Powers 225-coordinate endpoint has 7,867 terms and height 128,446, so it retains a support advantage; the new endpoints improve its variable, residual and height costs, not every cost simultaneously.

What is new, and what is credited. The advance is a compatible composition and fully instantiated certificate, not a priority claim for its component lemmas. The incoming Strict Horizons source already has

$$t = 16 + h^2, \qquad F = t^t, \qquad R = F, \qquad q = 4z + 1,$$

and strict local ranges using only the q^t denominator. Source 158, *Pooled Powers*, supplies fixed-base pooling and Pell-difference coordinates. Source 159, *Quadratic Projection 234*, supplies common-radical projection and integer row reduction. Source 157, *Linear CRT*, already uses an odd-linear beta multiplier in a different bootstrap. We prove compatibility with the unchanged strict frame and the mod-four radix, including the preliminary negative-beta possibility.

5.1 The unchanged frame supports both power pools

Before interpreting powers, the retained signed frame, semantic anchor and pooled orientation constraints give canonical signed Pell pairs and $|A| > W^W$. A literal retained-row identity gives

$$W = 2 + \sum_{j=1}^{34} a_j^2.$$

Every actual base, index and output of all seventeen retained requests is one of these affine atoms. They include $t, F + t, x, y + 1, F, B$, the nine CRT representatives, N, the actual global derived label, $q, q + 1, BH$, and the fourteen pooled outputs. Hence every required square is $< W$, and $t \geq 16$ gives $W \geq 258$. In the dense 235 chart the identity is row 152 plus twice row 4 plus row 169; in the sparse 236 chart it is row 153 plus row 4. None of these rows is deleted. Actual y is stored_y-1; this offset is checked.

Here is the credited pooling argument, included to expose its hypotheses. For a nonzero base n, let X_i, Y_i be canonical signed Pell coordinates at index k_i, put $Z_i = AY_i$, and assume $n^2, m_i^2, k_i^2 < W$ for $r \leq 15$ requests. Write

$$T = 2An - n^2 - 1, \quad D_i = m_i - X_i + Z_i, \quad H_n = \sum_i Y_i(m_i - X_i), \quad S_n = \sum_i (D_i^2 - Y_i^2).$$

Two new coordinates suffice, with quadratic equations

$$K_n - 2H_n - (2A - n)L_n = 0, \qquad S_n - nK_n + L_n = 0. \tag{1}$$

Indeed,

$$\sum_i (D_i - nY_i)^2 = S_n - 2nH_n - T\sum_i Y_i^2,$$

so T divides the squared-error sum. Let $b = \lfloor \sqrt{W-1} \rfloor \geq 16$. Since $T \equiv -1 \pmod{n}$, negative Pell indices can be cleared by multiplication by n^b. Thus T divides

$$E = \sum_i [m_i n^b - n^{b+k_i}]^2, \quad 0 \leq E \leq 4rb^{4b} < W^{2b+1} \leq W^W < |T|.$$

The last bound follows from $|T| \geq 2|A||n| - n^2 - 1 \geq 2|A| - W > W^W$. Therefore $E = 0$. At $|n| \geq 2$, each $k_i \geq 0$ and $m_i = n^{k_i}$; a negative index would give the impossible integer identity $m_i n^{-k_i} = 1$. At $n = \pm 1$ the signed unit output is exact without a nonnegative-index claim. Conversely true powers give the divisibility needed to set $L_n = (S_n - 2nH_n)/T$ and then $K_n = 2H_n + (2A - n)L_n$.

Apply this theorem to the eleven requests at $q + 1$ and the three at q. The initial requests $F = t^t, N = B^x, q = (BH)^x$ remain individual. The current chart has just one old explicit modulus, T_+, because

$$T_q = T_+ - 2A + 2q + 1.$$

Deleting fourteen quotients and that one modulus, and adding four pooled coordinates, saves eleven variables and eleven residuals, not twelve. All old power rows, retained AY gates, norm rows and offsets are checked literally before any deletion.

5.2 The odd-linear beta bootstrap in the correct order

Replace $v_\beta = (2w + 1)^2$, $y = Fv_\beta$ by

$$y = F(2w + 1). \tag{2}$$

The monic multiplier contraction saves one variable and one residual. Its multiplier may initially be negative, so the parent's positive-beta argument cannot simply be quoted.

The retained power request first proves $F = t^t > 1$. Consequently y is a nonzero multiple of F. Since $q = 4z + 1$, both $q \neq 0$ and $|q + 1| \geq 2$ hold immediately. Interpret the $q + 1$ pool first: it gives nonnegative actual representatives, hence $H = 2 + \sum a_\nu \geq 2$. The retained index row $B\mu = C + F + 1$ has positive right-hand side, so $B \neq 0$ and $|BH| \geq 2$. The individual self-radix request gives $x \geq 0$ and $q = (BH)^x$.

If $x = 0$, the last endpoint

$$x = 1 + [1 + (t + 1)y]u_t$$

would make $1 + (t+1)y$ equal to 1 or -1. This would give $y = 0$ or $(t+1)y = -2$, both impossible for a nonzero multiple of $F > 1$ and $t \geq 16$. Thus $x \geq 1$ and $|q| \geq 2$. The q pool now forces its requested index $y + 1 \geq 0$. Hence $y \geq -1$, and its nonzero-multiple property strengthens this to $y \geq F > 0$.

The original endpoint inequalities can now be used: from $x = F + 1 + (y+1)u_0$ one gets $u_0 \geq 0$, then $x \geq F + 1$, $u_t \geq 1$, and

$$x \geq 2 + (t + 1)y \geq 2 + (t + 1)F.$$

The inherited signed short-lift argument authentically recovers $B = \binom{F+t}{t} > 0$, and then $q = (BH)^x > 0$. All strict ranges and local rows follow as in Strict Horizons. The boundary $y = F$ is still excluded *after* local decoding: it would give $S_0 = 0$ and $S_t = 1$, while positive weighted updates propagate zero. Thus the source's label-product and Horn-closure argument gives a finite C222 refutation.

5.3 Completeness and unchanged arithmetic budgets

The lcm-selector hypothesis remains valid: $\Lambda_t \le 4^t$ gives $F = t^t \ge 16^t \ge \Lambda_t^2$, and the selector requires a weak inequality. Its reverse direction covers every prime power of B, not only row-isolating primes. The diagonal label-polynomial bound is also unchanged. After equal nonunit factor counts $d \le t$ have been recovered, each corresponding coefficient lies in $[1, F]$, so its *difference* has magnitude $< F$, even if an individual coefficient is F. Least-degree reduction modulo F recovers the whole polynomial.

For completeness choose an even h in the cofinal family $t = 16 + h^2$, strictly beyond the finite proof length and every local parameter and local derived label. This is a witness choice, not a new restriction on arbitrary integer solutions. Then t is even and $v_2(F) \ge t > v_2(t!)$. The parent constructs arbitrarily large valid $y = Fa^2$ with a an odd multiple of the odd part of $t!$. Equation (2) allows the same values, using $w = (a^2-1)/2$; an old square parameter maps explicitly to $2w_{\text{old}}(w_{\text{old}}+1)$. The odd beta moduli permit an arbitrarily large even x. The binomial is odd, the prescribed representative parities make H odd, and hence $q = (BH)^x$ is an odd square, supplying $z = (q-1)/4$. The unchanged all-primes CRT, short tails, norm and canonical Pell completion supply the other witnesses. The converse of (1) supplies each new K_n, L_n. This establishes the reverse implication.

5.4 Exact maps, complete coefficients, and evidence

Fourteen integral Pell-difference shears and a retained-norm row subtraction reduce density. Six common-Hessian-radical directions then yield integral quadratic zero-set bijections: along a direction d, $R_i(u + sd) = R_i(u) + sL_i(d)$; a unit coordinate of d and a unit row slope solve one coordinate without increasing degree. The six directions, restorations and inverse identities are frozen in certificates. Each final row operation $R_i \leftarrow R_i + kR_j$ has the inverse with $-k$.

The semantic pooling/beta changes preserve *zero existence*; they are not advertised as bijections of every predecessor witness tuple. The later coordinate and row transformations do have exact inverses on zero sets. The independent checker imports no producer, checks the full frame, pooling identities, all inverses, every collected quartic coefficient, liveness, metrics and corruption rejection. Normal Python and Python `-O` both pass. Reconstruction uses frozen certificates and a fresh temporary directory, never an optimizer. The original arithmetic-parent archive, proofs, notices and optional parent replay are retained. Finite checks do not replace the unbounded mathematical proof or a transitive proof-assistant build.

6 An inverse correction composed with the explicit Pell trace

The supplied polynomial has ten independently strictly positive integer unknowns, exact total degree 232, and an evaluator with 85 live binary operations: 51 multiplications, 18 additions and 16 subtractions. Squares and fixed scalar multiplications are charged; all leaves are integers. The matched incoming *Trace Radical* graph has 87 operations at the same degree. This is an ordinary proof composition at the explicitly retained native/syntactic-refutation calibration, not a consistency proof, global minimum theorem, or new formal-kernel object.

6.1 Complete equation

The inputs are $J, F, \alpha, i, T, w, s, n, \eta, \zeta > 0$. The four fixed integers β, K_0, M_C, M_F are supplied as complete payloads and by a finite seed recipe. Every following symbol is a polynomial abbreviation,

not an additional unknown:

$$h=\beta J,\qquad q=h+1,\qquad C=q-F-\alpha,$$
$$X=qw,\qquad Y=qs,\qquad k=\eta+\zeta,\qquad c=kY+\eta,$$
$$A=Y(X+1)+1,\qquad \Delta=A^2-1,\qquad H=4A-5,$$
$$\ell=(XY)(kY),\qquad E=1+\ell(\ell+k),$$
$$R=[h(q-F)+\alpha](q^2-1)+(M_C+qM_F)J,$$
$$B=(K_0+w)C-F,\qquad U=ic^2,\qquad a=\Delta U^2E,$$
$$F_0=1+a,\qquad G=EF_0,\qquad f=a+F_0,\qquad Q=4\Delta aF_0,$$
$$b=cT-R,\qquad V=fb-c,\qquad L=bV,\qquad K=X(3c-2X)+2,$$
$$e_6=6(k-R-1)-\ell(k^2-4),\qquad D_6=6q^6,\qquad M=hD_6H,$$
$$Z=hD_6[H(n-LB)+K]-H[D_6B+h(e_6-6q^4s)].$$

The single ordinary integer polynomial equation is

$$\boxed{P=Q(Z^2-M^2GV^2)+(M^2G-Z^2)=0.}$$

There are no division gates, variable powers, binomial primitives or hidden quantified variables. For proof only put

$$v=XY^2,\quad e_*=k-R-1-\frac{vk(k^2-4)}{6},\quad \rho=\frac{e_*}{q^6},\quad \sigma=\frac{s}{q^2},\quad \kappa=\frac{K}{H},\quad t=\frac{B}{h},\quad W=1+hL.$$

Before assuming any divisibility, the literal carrier identity is

$$z:=Z/M=n+\kappa+\sigma-Wt-\rho. \tag{3}$$

6.2 Signed soundness and the corrected index

The inherited trace-linear root extraction uses

$$f^2=1+4aF_0,\quad Q=4\Delta^2U^2G,\quad \gcd(E,F_0)=1.$$

At a zero it yields $G=g^2$, $E=\tau^2$, $F_0=f_0^2$, integral $z=\varepsilon gy$, and $QV^2-(Q-1)y^2=1$. The pair $(f,I)=(1+2a,2Ug)$ satisfies $f^2-\Delta I^2=1$ and $c^2\mid I$. Moreover $f\equiv 1 \pmod c$ identically. Signed folding gives $c=\psi_A(p)$ with positive odd p, and $k=2\psi_{1+2v}(m)$ with $m\geq 2$. Thus $c>8XY^3\geq 8q^4$ and $K>0$.

The retained main-index-three exclusion gives $c>c_3=4\Delta+3$. The trace-linear auxiliary-index-three identities are

$$f(b-4\Delta f)=c-c_3\qquad\text{or}\qquad f(b+4\Delta f)=c+c_3.$$

Their right sides are strictly between 0 and $2c<f$; hence the auxiliary index is at least five. The consequent bounds give $bV>0$, $gy<L/2$ and $|b|/q^6<L/2$. Because $k\geq 2$, the inverse correction is nonnegative before its subtraction: $e_*\leq k-R-1<b$, regardless of the sign of R. If $B\leq -1$, then $Wt<-L$ and (3) forces $n<L/2-L+|b|/q^6<0$. Therefore $B\geq 0$ and $C>0$. Only now use the calibrated mask estimate $3q<R<q^4-2$. The residues $V\equiv -c \pmod f$ and $V\equiv -R \pmod c$ give $p=R$; then R is odd and q is even.

The unbounded Lucas expansion is

$$k = 2m + 8vC_m + 32v^2 Z_m, \qquad C_m = \binom{m+1}{3}, \quad Z_m \geq 0.$$

Since k is even, $k(k^2-4)/6$ and e_* are integral. Reduce the literal Z modulo D_6, using $Z = Mz$; cancel the common ordinary integer factor 6, not a residue-ring unit. Since $\gcd(q, hH) = 1$, this proves

$$q^6 \mid e_* - q^4 s, \qquad q^4 \mid e_*.$$

For $t_0 = 2m$, $d = 2m - R - 1$, $A_m = C_m + 4vZ_m$ and $S_m = k^2 + kt_0 + t_0^2 - 4$, the exact cubic difference identity gives

$$3(e_* - d) = 4v^2(24Z_m - A_m S_m).$$

As $v = q^3 w s^2$, every prime of q gives $\operatorname{ord}_p(e_* - d) \geq 6\operatorname{ord}_p(q) - \operatorname{ord}_p(3) \geq 4\operatorname{ord}_p(q)$. Monotonicity yields $m < R$; consequently $|d| < R < q^4$ and $d = 0$. It is not yet legitimate to infer $q^6 \mid e_*$.

6.3 Project before recovering the full denominator

Reduction modulo H gives $H \mid 6hK$. Write $R = 2r + 1$. The authenticated ratio theorem gives $r \geq 48$, $Y > X^r/3$ and $X^{r+1} < H$. Universal projection yields

$$H \mid 6h(X - 2^R)(2^R X + 1).$$

With $u = 2^R = 2 \cdot 4^r$, the whole product satisfies

$$6h|(X-u)(uX+1)| < X^4(X+u)(uX+1) \leq 4 \cdot 4^r X^6 + 8 \cdot 16^r X^5 < X^{r+1} < H.$$

The last strict inequality follows at the minimum $(X, r) = (32, 48)$ from $2^{-117} + 2^{-25} < 1$, and improves thereafter. Thus $X = 2^R$, $H \mid K$, and q is a power of two. Now 3 is coprime to q, so the corrected-index identity proves $q^6 \mid e_*$. The carrier then gives $q^2 \mid s$ and $h \mid B$. Hence $q^3 \mid Y$ and $t \geq 0$ is integral, exactly the retained native reconstruction interface at source base $Y(X+1) + 2$.

6.4 Strictly positive completeness

A finite refutation supplies fresh canonical outer data. Choose a positive Pell period seed with $c^2\tau \mid I_0$ and set $i = I_0/(c^2\tau)$. The inherited positive auxiliary pair can be kept while replacing its old input by

$$T = \frac{V + c + Rf}{cf} = fT_{\text{old}} - \frac{f-1}{c}R > 0.$$

It is integral because $c \mid f - 1$ and satisfies $V = f(cT - R) - c$. The inverse correction can make ρ negative, but the explicit bounds

$$|\rho| < vk^3/3 < g/3, \qquad \kappa + \sigma < g/2$$

show that

$$n = gy + (1 + hL)t + \rho - \kappa - \sigma$$

is a strictly positive integer. The carrier is $z = gy$, proving $P = 0$. Both existential directions compose with the retained calibrated syntactic-refutation theorem; negation gives the equivalence with Con(ZFC).

6.5 Exact cost, coefficients and reproducibility

Deleting $v = (XY)Y$, sharing $\ell = (XY)(kY)$, and replacing the rank cubic by k^2, $k^2 - 4$, $\ell(k^2 - 4)$ saves two multiplications. For $q_0 = \beta J$, $k_0 = \eta + \zeta$ and $B_0 = k_0 sT - q_0^2(q_0 - F)$, the complete leading component is

$$[P]_{232} = -9 \cdot 2^{10} q_0^{100} w^{26} s^{58} i^{10} k_0^{32} B_0^2.$$

It is nonzero, not merely nonzero along a tested ray. The standard-library checker verifies 37 exact cuts covering all 85 gates, all 202 leading monomials, the full coefficient bytes and 171 rejected operator/missing-correction mutations. Normal and optimized receipts agree; finite recurrence diagnostics are separately labelled.

The principal tuple retains the incoming radix67 calibration. A second complete specialization uses the upstream one-anchor7309 tuple, credited rather than newly claimed: 172011 minimal raw bytes and largest port 345498 bits. Its complete native proof and finite Pc1 certificate are replayed with the new tuple; no old controller is substituted. The specialization is explicitly big-endian, unlike the principal little-endian payloads. All four ports are replaced together.

6.6 A small admissible-denominator interpolation

There is also a separate 77-operation, degree-348 graph in `q4_interpolation/`. It composes the retained q^4-denominator parent with the same unsquared trace; no new index correction is needed. Use the preceding definitions with $Y = q^3 s$, $D = q^4$, $e = k - R - 1$ and

$$M = hDH, \qquad Z = hD[H(n - bVB) + K] - H[DB + he].$$

The final polynomial is unchanged in form. Here $q^4 \mid e$ authenticates $2m = R + 1$; the full recurrence gives $4XY^2 \mid e$, and $q^4 \mid 4XY^2$ makes $\rho = e/q^4$ integral. The trace-linear signed bootstrap, projection and positive completeness then apply with $\sigma = 0$ and $\rho \geq 0$. The complete leading form is

$$[P]_{348} = -2^8 q_0^{216} w^{26} s^{60} i^{10} k_0^{34} T^2.$$

The independent checker verifies all 77 gates through 33 exact cuts and rejects 154 operator mutations. This is a composition/interpolation between the incoming 76/352 and 79/290 points, not a priority claim.

7 Verification and inherited calibration

The direct quartics are sums of squares of complete lists of quadratic integer residuals. Their expanded polynomials are supplied coefficient by coefficient. The compact equations are acyclic polynomial circuits over exactly the listed free inputs and complete fixed numerals. Powers, factorials, inequalities, divisions, Pell recurrences and binomial expressions occurring in proofs are not hidden primitives of a final polynomial.

Ordinary mathematical proofs and finite verification have distinct roles. The former establish unbounded soundness and completeness, including signs, denominator integrality, cofinal choices and arithmetic bootstraps. The latter bind those arguments to literal rows, gates, coefficient bytes, coordinate maps and full highest homogeneous components. Tests at finitely many points are not substituted for a universal argument.

The release authenticates its complete inventory and replays its checks in fresh disposable directories in normal and optimized Python. It checks exact identities, complete expansions, source pins, coefficient recipes, actual corruption controls and preservation of sealed bytes. The extracted final

ZIP is replayed separately before delivery. Detailed receipts state which commands were actually run; inherited historical receipts are not represented as fresh execution of all formal dependencies.

The direct constructions retain the numerical proof calculus and its ZF/ZFC refutation calibration. The compact and root branches retain their own specified native calibrations. A coherent alternate coefficient tuple usually changes the specialized polynomial. Similar port names, byte sizes or elementary numerical bounds never license mixing coefficients from different encodings. Storage endianness is recorded explicitly.

The reviews were performed separately within this AI-assisted research workflow, not by outside human referees. No upstream repository was changed. No new end-to-end Lean or Rocq replay, encoded object-language ZFC proof, literature-wide priority, or global minimality is asserted.

A The full original radix67 coefficient tuple

In this appendix K denotes the port K_0 in the new equations. Every symbol introduced here is a fixed constant, independent of the ten quantified variables. The formulas and 436-character seed are the retained indexed BFS943 radix67 specification; they reconstruct the exact supplied payloads, with $\beta = 2^{590069} - 1$.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} &= \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ &\quad (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ &\quad (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$h(q,b) = 2(q+1)+b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\},$$
$$\mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\},$$
$$r_q = 6 + |\mathcal{R} \cap \{0,\dots,q-1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0,\dots,q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w,n,e,s)$
$0, 1$	$(2,435,3,1),\ (5,0,0,0)$
$j+2\ (j = 2q+b < 430)$	$((1-d_j)l_{u_j}, h(q,b), d_j r_{u_j}, v_j)$
$432+b\ (b=0,1)$	$(0, h(215,b), 0, h(215,b))$
$434+b\ (b=0,1)$	$(0,b,0,b)$
$424+2r_q+b\ (q\in\mathcal{R}, b=0,1)$	$(r_q, b, 0, h(q,b))$
$424+2l_q+b\ (q\in\mathcal{L}, b=0,1)$	$(0, b, l_q, h(q,b))$
$924+j\ (0 \le j < 4)$	$(0, j+430\lfloor j/2\rfloor, 0, 438)$
928	$(2,434,2,0)$
929	$(3,436,4,4)$
930	$(4,436,4,0)$
$497+t\ (t=434,435,436)$	$(1,438,1,t)$
$\substack{934+3a+b-6\lfloor a/4\rfloor \\ a\in\{0,1,4\},\ b=0,1,2}$	$(a,437,b,437)$

For $t = (w,n,e,s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\,2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b\rfloor$. For $0 \le q < 215$, $b = 0,1$, $j = 2q+b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N,j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430+q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1-m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1+\sigma_j) + (1-\nu_j) D_{216}(T_0, j-\sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1,1,1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0,\dots,61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0,\dots,e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFot16F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

B The full one anchor7309 coefficient tuple

Here K denotes the fixed port K_0, not an unknown. The entire tuple is from the credited upstream one-anchor7309 construction and uses minimal unsigned big-endian payloads. The source mask shift occurs exactly once.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{7309}, \quad \beta = B - 1,$$
$$G_m(x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{4161} + \rho^{5228}.$$

$$\begin{aligned} M_C &= \beta + \rho - (\rho+3)\rho^{1063} \\ &\quad - G_{943}(\rho) - \Psi(\rho, 1), \\ M_F &= B + 3 + (2^{46} - 2)\Lambda \\ &\quad + \rho^{1013} G_9(\rho) + \rho^{1065} G_8(\rho) + \rho^{2079} \\ &\quad + 2^{25}\rho^{6343} - \rho^{6325}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1137} G_{943}(\rho) + \rho^{7351} \\ &\quad + \Lambda\Phi(1/\rho), \end{aligned}$$

where

$$\begin{aligned} \Phi(x) &= \sum_{i=0}^{942} [1 + 2^{10}(1 + w_i + 2^8 n_i \\ &\quad + 2^{17} p_i + 2^{25} s_i)] x^i + 2^{10}\Psi(x, 2), \\ \Psi(x, t) &= x^{1003}(x^{20} + t^{17}) G_8(tx) \\ &\quad + t^8 x^{983}(x^{30} + t^{17}) G_9(tx) \\ &\quad + t^{34}(2t - 1) x^{2079}. \end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j) r_{0,u_j}, h(q, b), d_j r_{1,u_j}, v_j)$
$434 - 2a + b\ (a, b = 0, 1)$	$(0, 432a + b, 0, 432a + b)$
$\substack{424 + 2r_{d,q} + b \\ q \in \mathcal{S}_d,\ d,b = 0,1}$	$(dr_{d,q}, b, (1 - d) r_{d,q}, h(q, b))$
$924 + 2a + b\ (a, b = 0, 1)$	$(0, 432a + b, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934 + 3a + b\ (a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(E_0, \tau_q) - 14),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j)D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```